

INFOSYS SPRINGBOARD

Documentation

Milestone-2

Name:- Soham Badgujar

Project Title:- Interactive Analytics Dashboard for Global Income Distribution

Project Statement and Outcomes:-

The Interactive Analytics Dashboard for Global Income Distribution project aims to develop an interactive Power BI dashboard that visualizes income inequality data across various countries and regions. This project involves collecting, preprocessing, and structuring global income distribution and economic data to enable effective visualization. The dashboard will provide users with insights into income disparities over time, country-wise comparisons, and patterns of inequality globally.

The outcome will be a user-friendly dashboard, embedded within a Streamlit web application, offering an intuitive interface for economists, researchers, policymakers, and the general public. This platform will help users explore global income inequality, understand disparities across different regions, and analyze trends in wealth distribution. The project will conclude with comprehensive testing to ensure dashboard accuracy and usability, supported by detailed documentation for future reference and enhancements.

Transformation creation

Measures Created:-

1. Total Updated Population

DAX formula:-

Total Updated Population
= SUM(fixed_global_income_distribution[Total_Updated_Population])

2. Total Countries

DAX formula-

Total Countries =
DISTINCTCOUNT(fixed_global_income_distribution[Country_Name])

3. Average Gini Index

DAX formula-

Avg Gini Index =
AVERAGE(fixed_global_income_distribution[Avg_Gini_Index])

4. Average Inequality Index

DAX formula-

Avg Inequality Index =
AVERAGE(fixed_global_income_distribution[Avg_Inequality_Index])

5. Latest Inequality

DAX formula-

Latest Inequality
= MAX(fixed_global_income_distribution[Latest_Inequality])

6. Palma Ratio Total

DAX formula-

Total Palma Ratio
= SUM(fixed_global_income_distribution[Palma_Ratio])

Calculated Columns Created:-

Inequality Category =
SWITCH(
 TRUE(),
 [Gini_Index_(World Bank)] < 30, "Low Inequality",
 [Gini_Index_(World Bank)] < 40, "Medium Inequality",
 [Gini_Index_(World Bank)] < 50, "High Inequality",
 "Very High Inequality"
)

Visualizations Used in Dashboard

1. KPI Cards

Purpose-

Shows key summary metrics

Cards Used

- Latest Inequality
- Avg Gini Index
- Avg Inequality Index
- Total Countries
- Total Updated Population

Fields Used

Values → Measures created

2. Waterfall Chart

Title-

Country Distribution by Income Group

Purpose-

Shows number of countries in each income classification

Fields Used-

Category → Income_Classification

Values → Country_Name (Count)

3. Clustered Column Chart

Title-

Richest 20% Income Share by Region

Purpose-

Shows income share across regions

Fields Used-

X-Axis → World_Bank_Region

Y-Axis → Income_Share: Richest_20%

4 Donut Chart

Title-

Palma Ratio by World Bank Region

Purpose-

Shows inequality ratio distribution

Fields Used-

Legend → World_Bank_Region

Values → Palma_Ratio

5 Bar Chart

Title-

Top 5 Countries by Inequality

Purpose-

Shows countries with highest inequality

Fields Used-

Y-Axis → Country_Name

X-Axis → Gini_Index_(World Bank)

Filter → Top 5

6 Line Chart

Title-

Global Gini Index Trend

Purpose-

Shows inequality trend over time

Fields Used-

X-Axis → Year

Y-Axis → Avg_Gini_Index

7 Column Chart

Title-

Categories by Inequality Level

Purpose-

Shows number of countries per inequality category

Fields Used-

X-Axis → Inequality_Category

Y-Axis → Country_Name (Count)

8. Slicers Used

Slicers allow filtering of dashboard.

Slicers added:

- Income Classification
- Inequality Category
- Last Census Year
- World Bank Region

9. Dashboard Design Features

Theme Used-

Dark Blue Professional Theme

UI Features-

- Glow Effects
- Rounded Cards
- Modern Visual Layout
- Interactive Filtering

10. Key Insights from Dashboard

- Global average Gini Index is 37.52
- Total countries analyzed: 200
- Total population analyzed: 7.85 Billion
- Namibia and South Africa have highest inequality
- Most countries fall under Medium and Low inequality

11. Tools Used

- Power BI Desktop
- DAX (Data Analysis Expressions)
- Data Visualization
- Data Modeling

11. Project Outcome

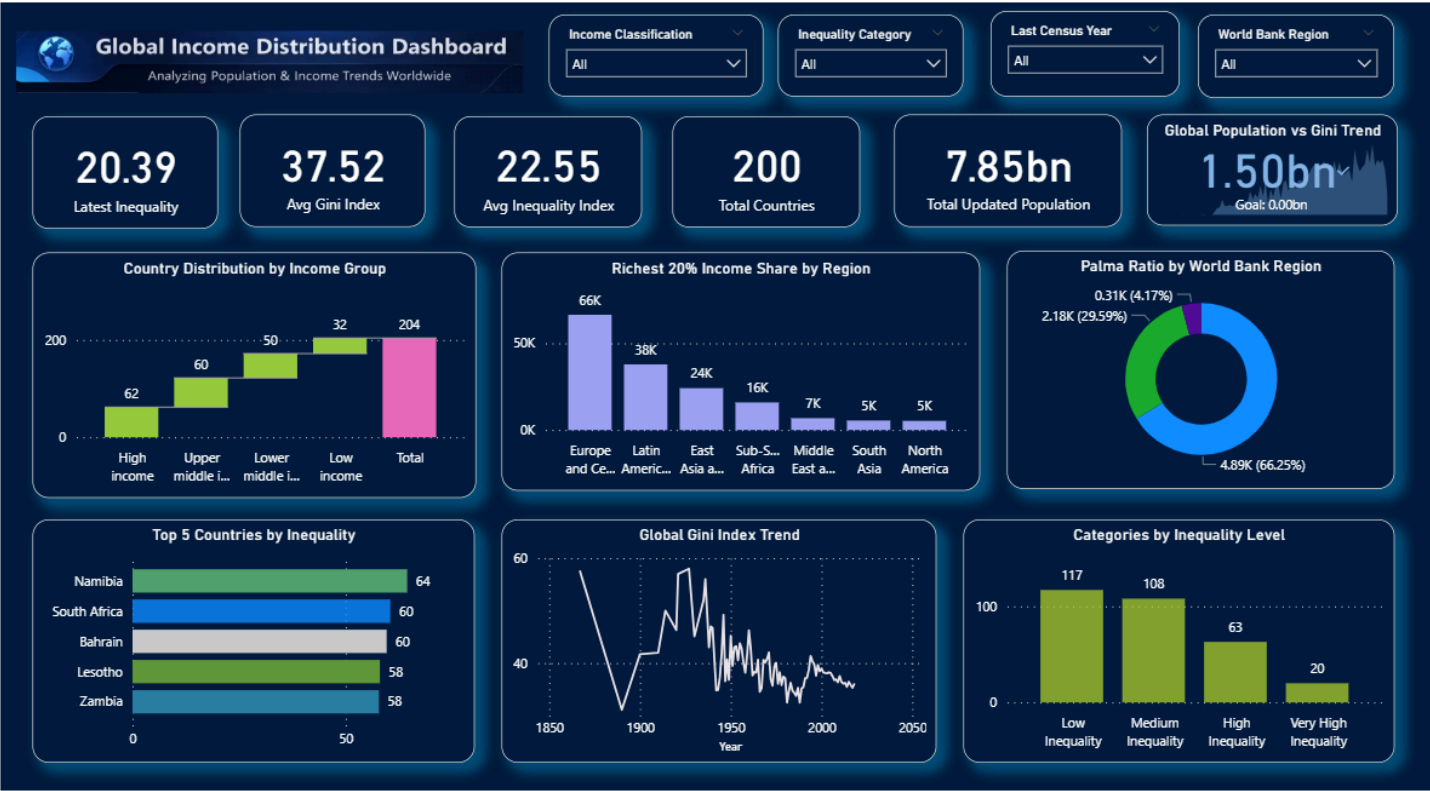
This dashboard helps:

- Identify inequality trends globally
- Compare income distribution across regions
- Analyze population and income share patterns
- Support economic and policy analysis

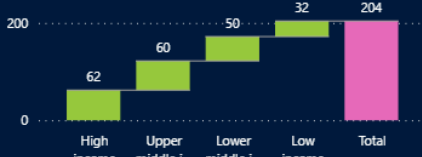
12. Summary-

The Global Income Distribution Dashboard analyzes income inequality, population distribution, and economic classification across countries and regions using Power BI. It highlights key metrics such as Gini Index, Inequality Index, Palma Ratio, and Total Population through interactive KPI cards and visualizations. The dashboard enables users to identify inequality trends, compare regions, and understand global income patterns. This project demonstrates effective use of data modeling, DAX measures, and interactive dashboard design for meaningful insights.

Dashboard-

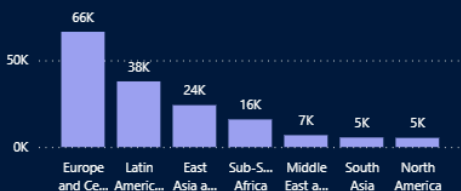


Country Distribution by Income Group



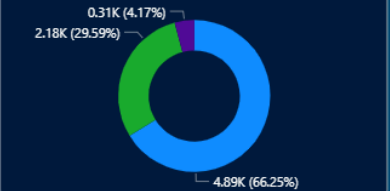
Income Group	Count
High income	62
Upper middle i...	60
Lower middle i...	50
Low income	32
Total	204

Richest 20% Income Share by Region



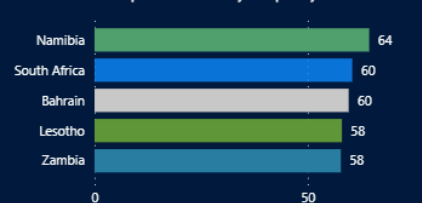
Region	Share
Europe and Ce...	66K
Latin Americ...	38K
East Asia a...	24K
Sub-S... Africa	16K
Middle East a...	7K
South Asia	5K
North America	5K

Palma Ratio by World Bank Region



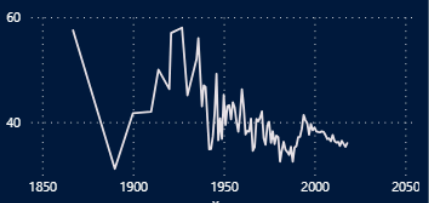
Region	Ratio	Percentage
Region 1	2.18K	29.59%
Region 2	0.31K	4.17%
Region 3	4.89K	66.25%

Top 5 Countries by Inequality



Country	Inequality Score
Namibia	64
South Africa	60
Bahrain	60
Lesotho	58
Zambia	58

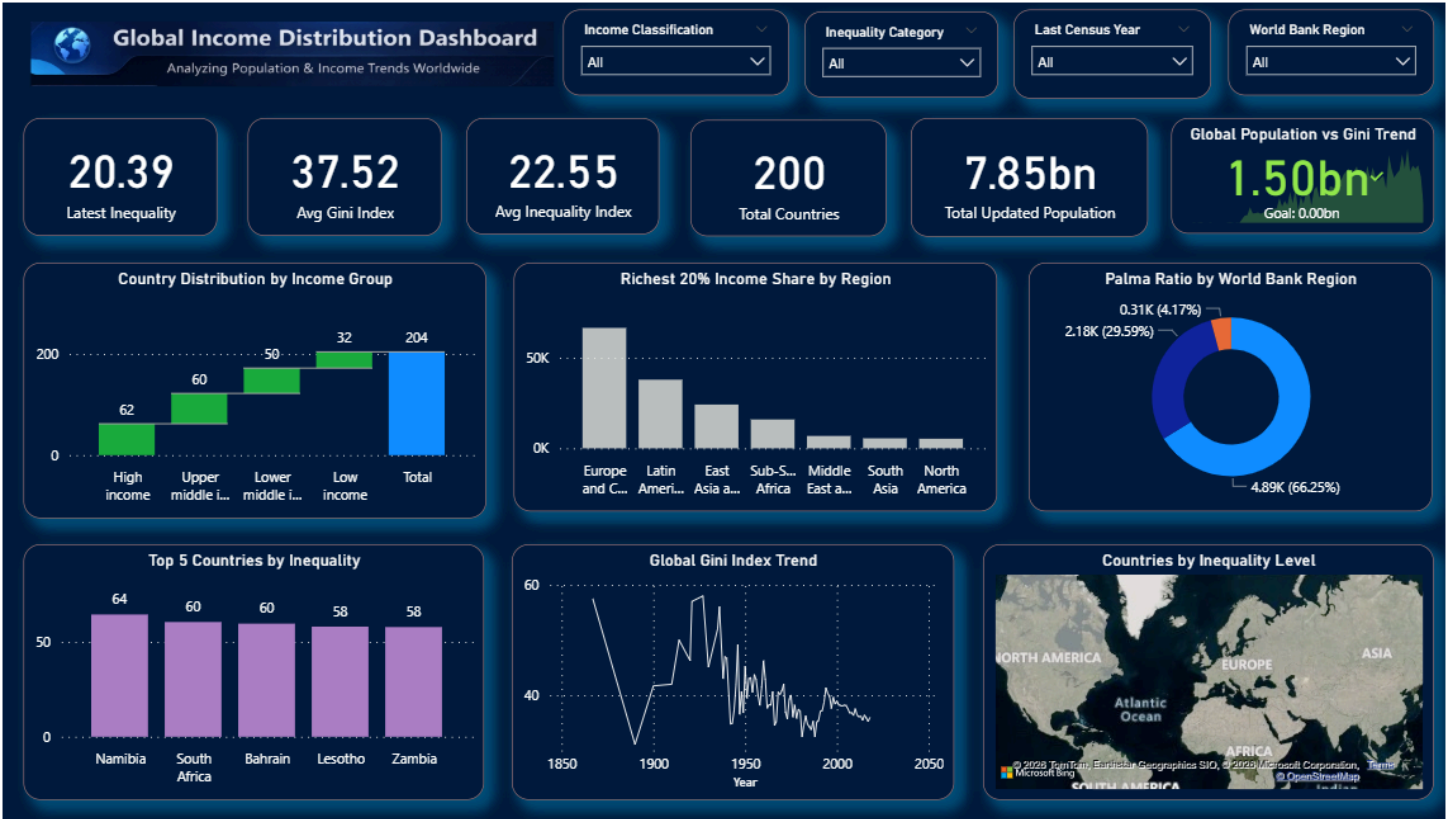
Global Gini Index Trend



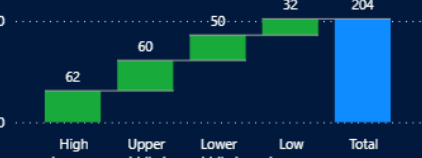
Categories by Inequality Level



Inequality Level	Count
Low Inequality	117
Medium Inequality	108
High Inequality	63
Very High Inequality	20

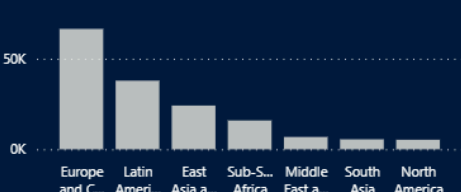


Country Distribution by Income Group



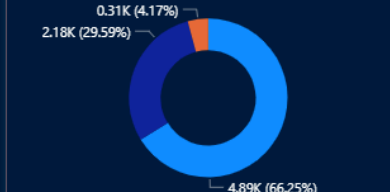
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Global Gini Index Trend



Countries by Inequality Level



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